Prayush B Menon

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OBJECTIVE

Embedded Systems & Firmware Engineer with an ECE background and hands-on expertise in bare-metal C development on PIC microcontrollers. Proficient in serial communication protocols (CAN, UART, SPI, I2C), Linux systems programming (POSIX, IPC, signals), memory-efficient data structures, and GDB debugging. Experienced in low-level firmware architecture, hardware-software integration, and software test automation.

TECHNICAL SKILLS

Languages: C / C++, Java
Microcontrollers & Protocols: PIC18F4580, CAN, UART, I2C, SPI
Systems & Concepts: POSIX System Calls, IPC (Pipes, Signals), Multiprocessing, Memory Management, Bit Manipulation
Developer Tools & OS: Linux (Fedora/Ubuntu), GDB, Make, Git, Neovim, Vim, Postman, Burp Suite
Software Testing & Management: Selenium, Playwright, Performance Monitoring (Lighthouse), Test Automation, JIRA

PROJECTS

Minishell (Unix Command Shell) GitHub

C, Linux Systems Programming, POSIX APIs, Signals, Process Management

- Built a Unix command shell in C supporting command execution, dynamic prompts, and multi-stage pipelines using `pipe()` and `dup2()`.
- Implemented signal handling for `SIGINT`, `SIGTSTP`, and `SIGCHLD`, with a linked-list job table to manage foreground and background processes.
- Handled built-in commands (`cd`, `pwd`, `exit`, `echo`), environment variable expansion (`$PATH`, `$?`), and asynchronous zombie process reaping.

Multithreaded TFTP Server & Client GitHub

C, UDP Sockets, POSIX Threads, Semaphores, Network Protocols

- Built a multithreaded TFTP file transfer server and client in C over UDP sockets, handling concurrent client uploads and downloads with POSIX threads.
- Limited active client transfers to 10 using a counting semaphore (`sem_trywait`) that sends instant busy replies, assigning each transfer to a dedicated worker thread and port.
- Implemented packet parsing for RRQ, WRQ, DATA, ACK, and ERROR opcodes with stop-and-wait flow control and binary octet transfer mode.

Concurrent HTTP/1.1 Server GitHub

C, POSIX Sockets, pthreads, Semaphores, HTTP/1.1, Systems Programming

- Architected a concurrent HTTP/1.1 web server in C using a thread pool of 8 worker threads and a bounded circular queue synchronized via mutexes and counting semaphores.
- Implemented an HTTP/1.1 protocol parser with dynamic URL routing (`/echo/{str}`), bounded header parsing (`User-Agent`, `Content-Length`), and strict socket lifecycle management.
- Engineered static file serving with automatic MIME-type detection for in-browser rendering, along with disk-backed binary file upload handling (`POST /files/`) returning `201 Created`.

Car Black Box (Event Data Recorder) GitHub

Embedded C, PIC18F4580, I2C, UART, ADC, EEPROM, RTC, CLCD

- Wrote bare-metal I2C functions for a PIC18F4580 microcontroller to log gear shifts, clock timestamps, and potentiometer values through ADC as speed values to EEPROM.
- Implemented a circular buffer to store and retrieve the 10 latest events, with a function to read matrix keypad values, a 16x2 CLCD menu, and UART data export to PC.
- Engineered secure password authentication with attempt counters, lockout timers, and non-volatile credential updates directly in EEPROM.

Car Dashboard (Automotive CAN Simulator) GitHub

Embedded C, PIC18F4580, CAN Protocol, LCD, Embedded Systems

- Built a two-node automotive dashboard simulator using PIC18F4580 microcontrollers communicating over a CAN bus.
- Transmitted real-time vehicle telemetry (speed, RPM, gear) from the engine node and decoded incoming CAN frames to update a live LCD dashboard.
- Configured CAN bit timings, standard 11-bit identifiers, and acceptance filter masks to isolate and prioritize critical sensor data packets.

Lexical Analyzer for C GitHub

C, Lexical Analysis, Tokenization, Binary Search, File I/O

- Built a C source code tokenizer that strips comments and preprocessor directives, classifying code into keywords, identifiers, constants, and operators.
- Used binary search over sorted keyword arrays for fast lookups, and added syntax checks for unclosed quotes and unmatched parentheses or braces.
- Engineered line and column tracking data structures to report descriptive compiler-style lexical diagnostics with exact error location indicators.

Red-Black Tree (Self-Balancing BST) GitHub

C, Data Structures, Algorithms, Tree Rotations, Memory Management

- Engineered a self-balancing Red-Black Binary Search Tree in C based on CLRS algorithms, guaranteeing $O(\log N)$ worst-case time complexity for search, insertion, and deletion.
- Implemented left and right tree rotations with recoloring fixups for insertion and deletion, resolving uncle cases and double-black height violations.
- Designed an explicit sentinel (`NIL`) node architecture to eliminate null-pointer boundary checks, with in-order, pre-order, and post-order traversals and an interactive CLI.

Arbitrary Precision Calculator (APC) GitHub

C, Data Structures, Algorithms, Dynamic Memory Allocation

- Built an arbitrary-precision calculator in C using doubly linked lists to perform arithmetic on numbers of any size without 64-bit integer limits.
- Implemented addition, subtraction, multiplication, and division with manual carry and borrow logic across nodes, ensuring zero memory leaks.
- Optimized digit compression and signed arithmetic routines, adding division by repeated subtraction with division-by-zero protection.

Image Steganography (LSB) GitHub

C, Bit Manipulation, Binary File I/O, Image Processing

- Built a command-line tool in C to hide and extract text messages inside 24-bit BMP images using the Least Significant Bit (LSB) technique.
- Replaced the lowest bits of RGB pixel bytes with secret payload data while preserving the 54-byte BMP header to keep images visually unchanged.
- Embedded a 2-byte magic string and 32-bit payload length header into pixel data to verify authenticity and avoid buffer overruns on decoding.

MP3 Tag Reader & Editor GitHub

C, Binary File Parsing, ID3v2 Tags, File I/O

- Built a command-line tool in C to view and edit ID3v2.3 metadata (title, artist, album) directly inside MP3 files.
- Parsed binary frame headers, handled big-endian byte offsets, and updated tag text in-place without corrupting the audio bitstream.
- Implemented dynamic tag frame resizing and file duplication algorithms to safely modify metadata without truncating audio frames.

EDUCATION & TRAINING

Emertxe Information Technologies Bangalore, India

Advanced Embedded Systems Training (PIC18F4580, C/C++, Protocols, Linux)

Amrita School of Engineering Kollam, Kerala

B.Tech in Electronics and Communications Engineering (CGPA: 8.1/10)

COURSE WORK

Data Structures & Algorithms (DSA), Microcontrollers, Embedded Systems, Operating Systems, Computer Networks